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HARD DISK DRIVE DEVICE

Abstract

A hard disk drive device includes: a base plate including a bottom surface part and a side wall part extending in a direction orthogonal to the bottom surface part along an edge part of an entire periphery of the bottom surface part; a structure provided more inside than the side wall part and including a female screw part; a cover having a plate shape, fixed to the base plate such that a back surface faces the bottom surface part, and including a through hole at a position facing the structure; a screw inserted into the through hole and including a male screw part fitted with the female screw part; and a first sealing member disposed at least one of between the structure and the cover and between the cover and the fastening body, and formed in an annular shape.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority to Japanese Patent Application Number 2024-026474 filed on Feb. 26, 2024. The entire contents of the above-identified application are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The disclosure relates to a hard disk drive device.

BACKGROUND

[0003] A housing of the hard disk drive device includes a base plate and a cover. In the housing, a recording disk rotating at a high speed and a magnetic head floating by an airflow generated by rotation of the recording disk are disposed. A gap between the recording disk and the magnetic head is several nanometers to several tens of nanometers. Therefore, the cleanliness inside the housing needs to be kept high.

[0004] As a method of keeping the cleanliness inside the housing high, enhancement of the sealability of the housing is conceivable. For example, JP 2015-1990 A discloses a hard disk drive device having a gasket disposed between the base plate and the cover.

[0005] The cover is fastened to the base plate by a screw component with the gasket disposed between the cover and the base plate. Here, a fastening portion includes, other than an outer edge part of the base plate, a structure such as a shaft of a spindle motor and a pivot post disposed in a space inside the housing.

SUMMARY

[0006] FIG. **11** is a cross-sectional view of a known hard disk drive device **201**. A gasket **230** is disposed between a base plate **210** and a cover **220**. On the other hand, as illustrated in FIG. **12**, no gasket is disposed between the cover **220** fastened by a screw **240** and the fastening portion of a shaft **270** of a spindle motor. It is desired to apply such a structure for improving the sealability also to the fastening portion of the structure disposed in a space between the cover and the housing.

[0007] An object of the disclosure is to provide a hard disk drive device with a structure for improving sealability between a cover and a fastening portion of a structure disposed in a space inside a housing.

[0008] In order to solve the above problem, a hard disk drive device is provided, the hard disk drive device including: a base plate including a bottom surface part and a side wall part extending in a direction orthogonal to the bottom surface part along an edge part of an entire periphery of the bottom surface part; a structure provided more inside than the side wall part and including a female screw part; a cover having a plate shape, fixed to the base plate such that one surface faces the bottom surface part, and including a through hole at a position facing the structure; a fastening body inserted into the through hole and including a male screw part fitted with the female screw part; and a first sealing member disposed at least one of between the structure and the cover and between the cover and the fastening body, and formed in an annular shape.

[0009] According to the disclosure, it is possible to provide a hard disk drive device including a structure for improving sealability between a cover and a fastening portion of a structure disposed in a space inside a housing.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. **1** is a perspective view of a hard disk drive device **1**.

[0011] FIG. **2** is a cross-sectional view of the hard disk drive device **1**.

[0012] FIG. **3** is a partial cross-sectional view of the cover **20**.
[0013] FIG. **4** is a partial cross-sectional view of a spindle motor **3** of FIG. **2**.
[0014] FIG. **5** is an enlarged view of a V part of FIG. **2**.
[0015] FIG. **6** is an enlarged view of the V part of FIG. **2** in a modification.
[0016] FIG. **7** is an enlarged view of the V part of FIG. **2** in a modification.
[0017] FIG. **8** is an enlarged view of the V part of FIG. **2** in a modification.
[0018] FIG. **9** is an enlarged view of the V part of FIG. **2** in a modification.
[0019] FIG. **10** is an enlarged view of the V part of FIG. **2** in a modification.
[0020] FIG. **11** is a cross-sectional view of the known hard disk drive device **201**.
[0021] FIG. **12** is an enlarged view of a part XII of FIG. **11**.

DESCRIPTION OF EMBODIMENTS

[0022] Hereinafter, an embodiment of the disclosure will be described with reference to the drawings. However, while various technically preferable limitations for carrying out the disclosure are attached to the embodiment described below, the scope of the disclosure is not limited to the following embodiment and illustrated examples.

[0023] FIG. **1** is a perspective view illustrating the configuration of the hard disk drive device **1**. FIG. **2** is a cross-sectional view of the hard disk drive device **1**.

[0024] Here, as illustrated in FIG. **2** and the like, a direction parallel to a center axis of a shaft **70** described later is defined as an axial direction, a direction around the center axis of the shaft **70** is defined as a circumferential direction, and a direction perpendicular to the axial direction is defined as a radial direction. For the sake of description, the axial direction is defined as an up-down direction, where the cover **20** side is the up and a base plate **10** side is down.

Hard Disk Drive Device

[0025] The hard disk drive device **1** includes a housing **2**, the spindle motor **3**, a recording disk **4**, and a data reading/writing device **5**.

[0026] The housing **2** is a box-shaped case accommodating the spindle motor **3** and the like. The housing **2** includes the base plate **10**, the cover **20**, a gasket **30**, and a screw **40** having a male screw part. The base plate **10** has a bottomed box shape with one surface opened. The gasket **30** is provided between the base plate **10** and the cover **20**. The gasket **30** is provided over the entire periphery of a side wall part **12** along the side wall part **12** of the base plate **10** described later. The cover **20** is fastened to the base plate **10** using the screw **40** being a fastening body. Due to this, the base plate **10** and the cover **20** form the housing **2** having an internal space S. Note that the gasket **30** may be another sealing means such as an adhesive.

[0027] The internal space S of the housing **2** is filled with air and other gases. The internal space S accommodates the spindle motor **3**, the recording disk **4**, and the data reading/writing device **5**.

[0028] The spindle motor **3** rotatably supports a plurality of the recording disks **4**. Note that a detailed structure of the spindle motor **3** will be described later.

[0029] The plurality of recording disks **4** are provided and supported by the spindle motor **3** such that respective disk surfaces face one another. Gaps are formed between the respective recording disks **4**. In the present embodiment, six recording disks **4** are supported by the spindle motor **3**. Note that for the sake of description, in a state of being supported by the spindle motor **3**, an upper front surface of the recording disk **4** is referred to as a front surface, and a lower front surface is referred to as a back surface.

[0030] The data reading/writing device **5** records data into the recording disk **4** or reads data from the recording disk **4**. The data reading/writing device **5** includes a pivot post **5***a*, a swing arm **5***b*, a voice coil motor **5***c*, and a magnetic head **5***d*.

[0031] The pivot post **5***a* swingably supports a plurality of the swing arms **5***b*. The pivot post **5***a* is a member extending in the up-down direction. The pivot post **5***a* has a female screw part **5***e* on the upper surface. In the present embodiment, the pivot post **5***a* is integrated with the base plate **10**. Note that the pivot post **5***a* may be configured as a component separate from the base plate **10** and attached to the base plate **10**.

[0032] The swing arm **5***b* is swingably supported by the pivot post **5***a*. The swing arm **5***b* is disposed in a gap between the recording disks **4** and on the uppermost surface and the lowermost surface of the recording disk **4**. The number of swing arms **5***b* is larger by one than the number of recording disks **4**. In the present embodiment, seven swing arms **5***b* are supported by the pivot post.

[0033] The voice coil motor **5***c* swings the swing arm **5***b* around the pivot post **5***a* as a rotation center. The voice coil motor **5***c* swings the swing arm **5***b* in parallel with the front surface of the recording disk **4**. The voice coil motor **5***c* has a female screw part **5***f* on the upper surface of a magnet disposed on the uppermost surface.

[0034] The magnetic head **5***d* magnetizes the recording disk **4** and reads magnetism from the recording disk **4**. The magnetic head **5***d* is provided at a tip end of the swing arm **5***b*. One magnetic head **5***d* is provided for each of the front surface and the back surface of the plurality of recording disks **4**.

[0035] The magnetic head **5***d* is disposed in a ramp mechanism **6** provided at a position away from the recording disk **4**. The ramp mechanism **6** is a portion forming a retraction position of the magnetic head **5***d*. When the voice coil motor **5***c* is activated and the swing arm **5***b* swings, the magnetic head **5***d* moves from the ramp mechanism **6** to above the front surface or below the back surface of the recording disk **4**, or moves from above the recording disk **4** to the ramp mechanism **6**.

[0036] When the spindle motor **3** rotates, the recording disk **4** also rotates. When the swing arm **5***b* swings in this state, the magnetic head **5***d* moves from the ramp mechanism **6** to above the front surface or below the back surface of the rotating recording disk **4**. Then, the magnetic head **5***d* magnetizes the recording disk **4** and records data onto the recording disk **4**. The magnetic head **5***d* reads magnetism from the recording disk **4** and reads data recorded on the recording disk **4**. On the other hand, when the magnetic head **5***d* does not magnetize the recording disk **4** or the magnetic head **5***d* does not read magnetism from the recording disk **4**, the swing arm **5***b* swings, and the magnetic head **5***d* moves to the ramp mechanism **6** from above the front surface or below the back surface of the rotating recording disk **4**.

Base Plate

[0037] The base plate **10** includes a bottom surface part **11**, the side wall part **12**, and the pivot post **5***a*.

[0038] The bottom surface part **11** corresponds to a bottom surface in the base plate **10** having a bottomed box shape with one surface opened. The bottom surface part **11** is a member having a rectangular plate shape. The bottom surface part **11** is provided with the side wall part **12**, a through hole **13**, and the pivot post **5***a*.

[0039] The side wall part **12** corresponds to a wall of a side wall in the base plate **10** having a bottomed box shape with one surface opened. The side wall part **12** extends in a direction orthogonal from the bottom surface part **11** along an outer edge part of the entire periphery of the bottom surface part **11**. The side wall part **12** extends from the bottom surface part **11** by the same length uniformly and has a tip end surface **14** at a tip end.

[0040] The tip end surface **14** is a surface opposite to the side connected to the bottom surface part **11** of the side wall part **12**. The tip end surface **14** is formed over the entire periphery of the side wall part **12**. A protrusion part **15** and a female screw part **16** are formed on the tip end surface **14**.

[0041] The protrusion part **15** is formed on the tip end surface **14** over the entire periphery of the side wall part **12**. The protrusion part **15** is formed at an outer edge part of the tip end surface **14**. The protrusion part **15** is formed in a direction (parallel to the extending direction of the side wall part **12**) orthogonal to the bottom surface part **11**.

[0042] The female screw part **16** is a portion having the screw **40** inserted. The female screw part **16** is formed toward the bottom surface part **11** in parallel with the extending direction of the side wall part **12** extending from the tip end surface **14**. In the present embodiment, six female screw

parts **16** are formed on the tip end surface **14**.
[0043] The through hole **13** is a hole provided so as to extend through the bottom surface part **11**. The shaft **70** of the spindle motor **3** is inserted into the through hole **13**.
[0044] The pivot post **5***a* extends in a direction orthogonal from the bottom surface part **11**. The height from the bottom surface part **11** of the pivot post **5***a* is lower than the height of the side wall part **12**. The pivot post **5***a* is provided in a region more inside than the side wall part **12** of the bottom surface part **11**. An upper end surface of the pivot post **5***a* is provided with the female screw part **5***e*.
Cover
[0045] FIG. **3** is a partial cross-sectional view of the cover **20**. The cover **20** is a member closing the opened surface of the base plate **10**. The cover **20** includes a plate surface part **21**, a through hole **22**, and a recess part **23**.
[0046] The plate surface part **21** is a member having a rectangular plate shape. The plate surface part **21** has a back surface **24** facing the bottom surface part **11** and a front surface **25** opposing to the back surface **24**. Here, opposing means back to back, and the front surface **25** is a surface facing the back surface **24** in the plate surface part **21**.
[0047] The through hole **22** is a hole extending through the plate surface part **21**. In the present embodiment, six through holes **22** are provided near an outer edge of the plate surface part **21**, and three are provided more inside than the outer edge of the plate surface part **21**. Note that for the sake of description, the through hole **22** provided near the outer edge of the plate surface part **21** is referred to as a through hole **22***a*, and the through hole **22** provided more inside than the outer edge of the plate surface part **21** is referred to as a through hole **22***b*.
[0048] The through hole **22***a* is provided at a position facing the female screw part **16** in a state where the cover **20** is attached to the base plate **10**.
[0049] The through hole **22***b* is provided at a position facing a structure provided more inside than the side wall part **12** of the base plate **10** in a state where the cover **20** is attached to the base plate **10**. Here, the structure provided more inside than the side wall part **12** of the base plate **10** is the pivot post **5***a*, the voice coil motor **5***c*, and the spindle motor **3**. The through hole **22***b* is provided at a position facing the female screw part **5***e* provided on the upper end surface of the pivot post **5***a*, the female screw part **5***f* provided on the upper end surface of the voice coil motor **5***c*, and a female screw part **71** provided on the upper end surface of the shaft **70** of the spindle motor **3** in a state where the cover **20** is attached to the base plate **10**.
[0050] The recess part **23** is provided around the through hole **22***b*. The recess part **23** is a portion recessed from the front surface **25**. The depth from the front surface **25** of the recess part **23** is larger than the thickness of the plate surface part **21**. Therefore, the plate surface part **21** protrudes on the back surface **24** side at the portion provided with the recess part **23**. In the present embodiment, three recess parts **23** are provided in the plate surface part **21**. The recess part **23** has a first recess part **26** and a second recess part **27**.
[0051] The first recess part **26** accommodates the screw **40** in a portion recessed from the front surface **25**. The first recess part **26** is continuous in an up-down direction with the through hole **22***b*. The depth of the first recess part **26** is larger than an axial length of the head of the screw **40**. The first recess part **26** is provided in a annular shape in axial view around the through hole **22***b*. The first recess part **26** has an inner diameter gradually decreasing from the front surface **25** side toward the back surface **24** side. The minimum inner diameter of the first recess part **26** is larger than the diameter of the through hole **22***b* and larger than the diameter of the head of the screw **40**. A bottom surface part **28** coming into contact with the head of the screw **40** is formed at a lower end of the first recess part **26**.
[0052] The second recess part **27** is provided continuously in the axial direction with the first recess part **26**. The second recess part **27** is provided in a annular shape in axial view around the through hole **22***b*. The second recess part **27** has an outer diameter larger than the outer diameter of the first

recess part **26** in axial view.

Spindle Motor

[0053] Next, a detailed configuration of the spindle motor **3** will be described. FIG. **4** is a partial cross-sectional view of the spindle motor **3**. The spindle motor **3** includes a stationary part **50** and a rotation part **60** rotating with respect to the stationary part **50** via a bearing mechanism.

Stationary Part

[0054] The stationary part **50** includes the shaft **70** and a stator core **80**.

[0055] The shaft **70** is a metal member having a columnar rod shape. A portion partially including one end of the shaft **70** is inserted into the through hole **13** of the base plate **10** and fixed with an adhesive. The shaft **70** is provided with the female screw part **71** on the other end surface (upper end surface).

[0056] The stator core **80** is a member having a plurality of electromagnetic steel sheets having an annular shape in axial view stacked in the axial direction. The stator core **80** is disposed near the through hole **13** and is fixed to the base plate **10** by a method such as adhesion. The stator core **80** has a plurality of pole teeth (salient poles) extending radially outward and arranged along the circumferential direction. A coil **81** is wound around the pole teeth. The stator core **80** generates a magnetic flux when a current flows through the coil **81**.

Rotation Part

[0057] The rotation part **60** includes a rotor hub **90** and a rotor magnet **100**.

[0058] The rotor hub **90** is a member rotating with respect to the shaft **70**. The rotor hub **90** includes an inner cylindrical wall part **91**, a disc part **92**, an outer cylindrical wall part **93**, and an outer edge part **94**.

[0059] The inner cylindrical wall part **91** is a member having a substantially cylindrical shape. A rotor hub through hole **95** axially extending through the rotor hub **90** is formed at the center (a part corresponding to the rotation center of the rotor hub **90**) of the inner cylindrical wall part **91**. The shaft **70** is inserted into the rotor hub through hole **95**.

[0060] The disc part **92** is a member having a disc shape coaxially with the center of the inner cylindrical wall part **91** in axial view. The disc part **92** is formed radially outward from the inner cylindrical wall part **91**.

[0061] The outer cylindrical wall part **93** is a member having a cylindrical shape having a thickness in the radial direction. The outer cylindrical wall part **93** is provided coaxially with the center of the inner cylindrical wall part **91** in axial view, and protrudes axially downward. The outer cylindrical wall part **93** is provided at the outer edge of the disc part **92**.

[0062] The outer edge part **94** is a member having an annular shape. The outer edge part **94** is provided at the lower end of the outer cylindrical wall part **93**. The outer edge part **94** protrudes radially outward from the outer cylindrical wall part **93** and is formed in a flange shape. The plurality of recording disks **4** are installed above the outer edge part **94** and radially outside the outer cylindrical wall part **93**.

[0063] The plurality of recording disks **4** are installed so as to be stacked in the axial direction. Between two recording disks **4** axially adjacent to each other, a spacer **110** is disposed and a gap is formed. That is, the recording disks **4** and the spacers **110** are alternately stacked in the axial direction. The plurality of recording disks **4** and the spacers **110** stacked in this manner are fixed to the rotor hub **90** by a clamp **112** attached to the upper surface of the rotor hub **90** with a screw part **111**.

[0064] The rotor magnet **100** is a member having an annular shape having a magnetic pole structure magnetized in a state where the polarity is inverted to N, S, N, S . . . along the circumferential direction in axial view. The rotor magnet **100** is attached to the inner peripheral surface of the outer cylindrical wall part **93**.

[0065] A minute gap is formed between the shaft **70** and the inner peripheral surface of the rotor hub through hole **95**. Then, the minute gap is filled with lubricating oil. Furthermore, a dynamic pressure generation groove not illustrated is formed in at least one of the shaft **70** and the rotor hub **90**. Due to this, a fluid dynamic bearing is formed.

Operation of Spindle Motor

[0066] When the coil **81** is energized, the magnetic attraction force and the magnetic repulsive force generated between the magnetic pole of the rotor magnet **100** and the pole teeth of the stator core **80** are switched. As a result, the rotation part **60** rotates about the shaft **70**.

[0067] As the rotor hub **90** of the rotation part **60** rotates at a high speed, the lubricating oil filled in the minute gap is pressurized by the dynamic pressure generation groove. As a result, dynamic pressure is generated in the fluid dynamic bearing. Due to the generated dynamic pressure, the rotor hub **90** rotates while being supported in a non-contact state with respect to the shaft **70**.

Sealing Member

[0068] In the present embodiment, the hard disk drive device **1** includes a first sealing member **121** and a third sealing member **123** in order to enhance the sealability of the internal space S.

[0069] The first sealing member **121** seals a gap of disposed portions. The first sealing member **121** is disposed at least one of between the structure and the cover **20** and between the cover **20** and the screw **40**. The first sealing member **121** is formed in an annular shape. The first sealing member **121** surrounds the screw **40**. In the present embodiment, the first sealing member **121** is a solid gasket formed in the annular shape. The first sealing member **121** is disposed between the upper end surface of the pivot post **5***a* and the cover **20**, between the upper end surface of the voice coil motor **5***c* and the cover **20**, and between the upper end surface of the shaft **70** and the cover **20**. The first sealing member **121** is in contact with each of the upper end surface of the pivot post **5***a* and the back surface **24** of the cover **20**, the upper end surface of the voice coil motor **5***c* and the back surface **24** of the cover **20**, and the upper end surface of the shaft **70** and the back surface **24** of the cover **20**.

[0070] The third sealing member **123** seals the first recess part **26** from the front surface **25** side. The third sealing member **123** is provided in the second recess part **27** so as to cover the first recess part **26**. The third sealing member **123** has a circular shape in axial view. The outer diameter of the third sealing member **123** is larger than the maximum inner diameter of the first recess part **26**. The material of the third sealing member **123** is resin or metal. The third sealing member **123** is fixed to the plate surface part **21** by applying an adhesive or a pressure-sensitive adhesive to the outer edge part of the surface disposed on the first recess part **26** side, or by welding the outer edge part to the plate surface part **21**. In the present embodiment, the third sealing member **123** is a resin member having a circular shape in axial view. The third sealing member **123** is fixed to the plate surface part **21** by applying an adhesive to the outer edge part of the surface disposed on the first recess part **26** side.

[0071] Here, the first sealing member **121** is disposed at least one of between the structure provided more inside than the side wall part **12** of the base plate **10** and the cover **20** and between the cover **20** and the screw **40** inserted into the through hole **22***b*. The third sealing member **123** is provided in the second recess part **27** so as to cover the first recess part **26** continuous with the through hole **22***b*. That is, the first sealing member **121** and the third sealing member **123** are provided more inside than the gasket **30** provided over the entire periphery of the side wall part **12** along the side wall part **12** of the base plate **10**.

Modifications

[0072] Note that the hard disk drive device **1** may be a combination of the following changes.

(1) Modification 1

[0073] The first sealing member **121** may be provided at other than the positions illustrated in FIGS. **2** and **5**. At this time, the shape of the cover **20** can be any shape. As illustrated in FIG. **6**, for example, the cover **20** includes a protrusion part **29***a* formed so as to axially protrude from the lower surface of the bottom surface part **28**. The protrusion part **29***a* is in contact with the upper end surface of the shaft **70**. The outer diameter of the protrusion part **29***a* is smaller than the outer diameter of the shaft **70**. At this time, a gap is formed between the lower surface of the bottom surface part **28** and the upper end surface of the shaft **70**. The first sealing member **121** is disposed in the gap between the lower surface of the bottom surface part **28** and the upper end surface of the shaft **70**.

[0074] As illustrated in FIG. **7**, for example, the cover **20** includes a protrusion part **29***b* formed so as to axially protrude from the lower surface of the bottom surface part **28**, and a groove part **29***c* provided on the lower end surface of the protrusion part **29***b*. The protrusion part **29***b* is in contact with the upper end surface of the shaft **70**. The outer diameter of the protrusion part **29***b* is equal to the outer diameter of the shaft **70**. The lower end surface of the protrusion part **29***b* is provided with the groove part **29***c* formed so as to be recessed upward. The groove part **29***c* is provided in an annular shape on the lower end surface of the protrusion part **29***b*. The first sealing member **121** is disposed in the groove part **29***c*.

(2) Modification 2

[0075] As illustrated in FIG. **8**, the hard disk drive device **1** may have the first sealing member **121** disposed between the cover **20** and the screw **40**.

[0076] As illustrated in FIG. **9**, a second sealing member **122** may be further disposed between the screw **40** and the first sealing member **121**.

[0077] The second sealing member **122** reduces a force applied to the first sealing member **121** at the time of tightening the screw **40**. The second sealing member **122** is a member having an annular shape in axial view. The second sealing member **122** surrounds the screw **40**. The material of the second sealing member **122** is resin or metal. The second sealing member **122** is fixed to the upper side of the first sealing member **121** by applying an adhesive or a pressure-sensitive adhesive to a flat surface in contact with the first sealing member **121**. Note that the second sealing member **122** is provided more inside than the gasket **30** provided over the entire periphery of the side wall part **12** along the side wall part **12** of the base plate **10**.

(3) Modification 3

[0078] As illustrated in FIG. **10**, in the hard disk drive device **1**, the screw **40** may be applied with a fourth sealing member **124** in the first recess part **26** in a state where the screw **40** is inserted into the through hole **22***b*.

[0079] The fourth sealing member **124** seals a gap between the screw **40** and the bottom surface part **28**. The fourth sealing member **124** is applied to the screw **40** in the first recess part **26**. The fourth sealing member **124** is an adhesive or a liquid gasket. The fourth sealing member **124** is applied to the outer peripheral surface of the screw **40** after the screw **40** is fastened to the female screw part **5***e*, the female screw part **5***f*, and the female screw part **71**. Note that the liquid gasket is a formed in place gasket (FIPG), a cured in place gasket (CIPG), or the like. The fourth sealing member **124** is provided more inside than the gasket **30** provided over the entire periphery of the side wall part **12** along the side wall part **12** of the base plate **10**.

[0080] Note that in FIGS. **6** to **10**, the portion illustrated in FIG. **5** (the spindle motor **3** described as a structure having a female screw part) is applied with each modification. In each of the above-described modifications, the structure having the female screw part is also applied to the portions of the pivot post **5***a* and the voice coil motor **5***c*. The structure having the female screw part is not limited to the spindle motor **3**, the pivot post **5***a*, and the voice coil motor **5***c*, and may be another structure. Furthermore, the above-described configuration may also be applied to the portion of the through hole **22***a* of the cover **20** fastened to the female screw part **16** of the base plate **10** with the screw **40**.

Effects

[0081] (Aspect 1) The hard disk drive device **1** according to the present embodiment includes: the base plate **10** including the bottom surface part **11** and the side wall part **12** extending in the direction orthogonal to the bottom surface part **11** along the edge part of the entire periphery of the bottom surface part **11**; the structure provided more inside than the side wall part **12** and including the female screw parts **5***e*, **5***f*, and **71**; the cover **20** having a plate shape, fixed to the base plate **10** such that the back surface **24** faces the bottom surface part **11**, and including the through hole **22***b* at a position facing the structure; the screw **40** inserted into the through hole **22***b* and including the male screw part fitted with the female screw parts **5***e*, **5***f*, and **71**; and the first sealing member **121** disposed at least one of between the structure and the cover **20** and between the cover **20** and the screw **40**, and formed in an annular shape.

[0082] In the hard disk drive device **1** described above, the gap between the structure and the cover **20** or the gap between the cover **20** and the screw **40** is sealed by the first sealing member **121**. Therefore, air outside the housing **2** is less likely to flow into the internal space S through these gaps. That is, the sealability of the internal space S is improved. As described above, it is possible to provide a hard disk drive device including a structure for improving sealability between a cover and a fastening portion of a structure disposed in a space inside a housing.

[0083] (Aspect 2) In Aspect 1, the hard disk drive device **1** further includes the second sealing member **122** disposed between the screw **40** and the first sealing member **121**.

[0084] Since the hard disk drive device **1** described above further includes the second sealing member **122**, the gap between the cover **20** and the screw **40** is more easily sealed. Therefore, air outside the housing **2** is less likely to flow into the internal space S through the gap between the cover **20** and the screw **40**. That is, the sealability of the internal space S is further improved.

[0085] When the screw **40** is fastened, the second sealing member **122** disposed between the screw **40** and the first sealing member **121** functions as a cushioning material. As a result, the first sealing member **121** is less likely to be applied with an external force when the screw **40** is fastened, and the first sealing member **121** is less likely to be displaced.

[0086] (Aspect 3) In Aspect 1 or 2, the first sealing member **121** is a liquid gasket.

[0087] In the hard disk drive device **1** described above, since the first sealing member **121** is the liquid gasket, the first sealing member **121** easily enters the gap between the structure and the cover **20** or the gap between the cover **20** and the screw **40**. Therefore, air outside the housing **2** is less likely to flow into the internal space S through the gap between the structure and the cover **20** or the gap between the cover **20** and the screw **40**. That is, the sealability of the internal space S is further improved.

[0088] (Aspect 4) In any of Aspects 1 to 3, in the hard disk drive device **1**, the cover **20** further includes the front surface **25** opposing to the back surface **24** and the recess part **23** recessed from the front surface **25**, the recess part **23** includes the first recess part **26** provided around the through hole **22***b* and accommodating the screw **40** in a portion recessed from the front surface **25**, and the second recess part **27** provided continuously with the first recess part **26** and having an outer diameter larger than the outer diameter of the first recess part **26** in axial view of the through hole **22***b*, and the third sealing member **123** provided in the second recess part **27** so as to cover the first recess part **26** is further included.

[0089] In the hard disk drive device **1** described above, the through hole **22***b* is closed by the third sealing member **123** provided in the second recess part **27** so as to cover the first recess part **26**. Therefore, the through hole **22***b* is doubly sealed by the first sealing member **121** and the third sealing member **123**. Therefore, air outside the housing **2** is less likely to flow into the internal space S through the gap between the structure and the cover **20** or the gap between the cover **20** and the screw **40**. That is, the sealability of the internal space S is further improved.

[0090] (Aspect 5) In any of Aspects 1 to 4, the hard disk drive device **1** further includes the fourth sealing member **124** applied to the screw **40** in the first recess part **26**.

[0091] In the hard disk drive device **1** described above, the gap between the cover **20** and the screw **40** is easily sealed by the fourth sealing member **124**. Therefore, air outside the housing **2** is less likely to flow into the internal space S through the gap between the cover **20** and the screw **40**. That is, the sealability of the internal space S is further improved.

[0092] (Aspect 6) In Aspect 5, in the hard disk drive device **1**, the fourth sealing member **124** is a

liquid gasket.
[0093] In the hard disk drive device **1** described above, since the fourth sealing member **124** is a liquid gasket, the fourth sealing member **124** easily enters the gap between the cover **20** and the screw **40**. As a result, the gap between the cover **20** and the screw **40** is easily sealed by the fourth sealing member **124**. Therefore, air outside the housing **2** is less likely to flow into the internal space S through the gap between the cover **20** and the screw **40**. That is, the sealability of the internal space S is further improved.
[0094] While preferred embodiments of the disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the disclosure. The scope of the disclosure, therefore, is to be determined solely by the following claims.

Claims

1. A hard disk drive device comprising: a base plate including a bottom surface part and a side wall part extending in a direction orthogonal to the bottom surface part along an edge part of an entire periphery of the bottom surface part; a structure provided more inside than the side wall part and including a female screw part; a cover having a plate shape, fixed to the base plate such that one surface faces the bottom surface part, and including a through hole at a position facing the structure; a fastening body inserted into the through hole and including a male screw part fitted with the female screw part; and a first sealing member disposed at least one of between the structure and the cover and between the cover and the fastening body, and formed in an annular shape.
2. The hard disk drive device according to claim 1, further comprising: a gasket provided between the base plate and the cover and over the entire periphery of the side wall part along the side wall part.
3. The hard disk drive device according to claim 1, further comprising: a second sealing member disposed between the fastening body and the first sealing member.
4. The hard disk drive device according to claim 1, wherein the first sealing member is a liquid gasket.
5. The hard disk drive device according to claim 1, wherein the cover further includes an other surface opposing to the one surface and a recess part recessed from the other surface, the recess part includes a first recess part provided around the through hole and accommodating the fastening body in a portion recessed from the other surface, and a second recess part provided continuously with the first recess part and having an outer diameter larger than an outer diameter of the first recess part in axial view of the through hole, and a third sealing member provided in the second recess part so as to cover the first recess part is further included.
6. The hard disk drive device according to claim 5, further comprising: a fourth sealing member applied to the fastening body in the first recess part.
7. The hard disk drive device according to claim 6, wherein the fourth sealing member is a liquid gasket.